

Household Demography - ODD model description

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This document describes the Household Demography model following the Overview, Design concepts, and Details (ODD) protocol as updated by Grimm et al. (2020). It describes the model family itself, not the simulation experiments used to analyse it. The main article and accompanying analysis repository document experimental designs, parameter sampling, outcome classifications, sensitivity analyses, and statistical post-processing separately.

The ODD covers versions v1.1–v1.4.2. These versions share a common household architecture and demographic submodels but differ along two controlled design dimensions: resource-dependent mortality and temporal resolution. Version numbers therefore identify alternative formulations within one model family rather than successive software releases in the strict sense of semantic versioning.

1 Purpose and patterns

1.1 Purpose

The Household Demography model was developed to examine how population dynamics in small-scale and pre-industrial populations emerge from age- and sex-structured demographic events, stochasticity, and household organisation, and how those dynamics change when demographic regulation and the temporal scheduling of events are represented differently. Its higher-level purpose is explanatory and methodological rather than predictive: the model provides a common framework in which demographic and socioecological mechanisms can be varied independently and their consequences for population persistence, population structure, and household organisation compared.

The model addresses three related questions: how combinations of fertility, mortality, union formation, household organisation, and stochastic demographic events generate different population trajectories and structures; how resource limitation modifies those dynamics when demographic load depends on population size or age composition and productive capacity depends on available labour; and how annual versus daily scheduling changes the realised consequences of otherwise comparable demographic assumptions.

The model is intended for comparative studies in which demographic evidence is incomplete or does not justify calibration to a single reconstructed population.

It is also intended to function as a reusable demographic component of broader socioecological simulations in which population size and composition interact with subsistence processes. The present model therefore represents demographic and household processes explicitly while leaving production, food allocation, environmental variation, and other context-specific socioecological processes outside its boundary.

1.2 Patterns

The model was not calibrated to reproduce the demographic history of a specific population, and no single empirical dataset is used as an acceptance criterion. Instead, we evaluate its suitability for its intended comparative and explanatory purpose against qualitative patterns implied by demographic theory and the mechanisms the model is designed to represent. These patterns concern emergent population behaviour rather than exact numerical targets.

- P1. Demographic replacement responds to age-specific fertility and mortality.** Parameter combinations producing greater realised fertility or lower mortality generate stronger demographic replacement, reflected in greater persistence or population growth and in corresponding changes in population age structure. Conversely, combinations producing insufficient replacement increase the probability of decline and extinction.
- P2. Density-independent populations have no endogenous demographic equilibrium.** In the absence of resource-dependent mortality, populations with sufficient demographic replacement can continue to grow until an externally imposed technical stopping condition is reached, whereas populations with insufficient replacement decline or become extinct. The model architecture does not impose stable population size.
- P3. Resource-dependent mortality generates negative demographic feedback.** In density-regulated versions, increasing demographic load relative to available capacity increases mortality and thereby constrains population growth. Regulation is continuous rather than imposed as a hard population ceiling, so populations can occur below, near, or temporarily above nominal capacity.
- P4. Population composition affects demographic pressure when demand is age-weighted.** Populations of equal size, but different age structures can exert different total subsistence demand and therefore experience different levels of density-dependent mortality. Population head count therefore no longer determines demographic pressure in demand-weighted versions.
- P5. Population composition affects realisable capacity when labour is age-weighted.** In labour-weighted versions, populations with different age structures can realise different fractions of the same nominal environmental

capacity because their available productive labour differs. Demographic pressure therefore depends jointly on subsistence demand and the population's capacity to provide labour.

P6. Explicit reproductive timing changes realised demographic dynamics. Daily scheduling preserves annual event probabilities at the level of repeated independent trials while introducing gestation and postpartum amenorrhoea as explicit reproductive states. The resulting birth timing and population dynamics can therefore differ from those of annual versions using otherwise comparable demographic schedules.

These patterns define the qualitative behaviours used to evaluate whether the model family performs its intended role as a comparative demographic framework. They do not constitute validation of the model as a reconstruction of any particular past population.

2 Entities, state variables and scales

2.1 Entities

The model represents two entity types: households and individuals. Households are explicit NetLogo agents, whereas individuals are demographic records stored within households. In addition, the NetLogo observer maintains global model state used to coordinate demographic processes, resource regulation, observation, and time keeping. NetLogo patches do not constitute modelled environmental entities.

Households. Households are instances of the NetLogo `households` breed. Each household has a unique NetLogo agent identifier and stores its lineage history, household age, maximum number of co-resident couples, and the demographic state of its members. The household is therefore both the computational agent and the principal social unit through which co-residence, partnership, descent, post-marital residence, household continuity, and household fission are represented.

Individuals. Individuals are not independent NetLogo agents. Within each household, individual state is encoded in parallel lists whose common index identifies the same person across variables. All model versions use `hh_membersAge`, `hh_membersSex`, and `hh_membersMarriage` with the same common-index convention; daily versions additionally use `hh_membersGestation` and `hh_membersAmenorrhea` to represent reproductive state. The temporal variables retain the same representation across model versions but are expressed in years in annual versions and days in daily versions. This representation preserves individual demographic state and relationships while demographic procedures are executed through household agents.

Global model state and population boundary. The NetLogo observer maintains state shared by the simulated population, including demographic probability tables, temporary pools used during union formation and orphan reassignment, population and event counters, time counters, and, in resource-regulated versions, quantities used to calculate density-dependent mortality. The population boundary is conceptual rather than spatial. External partners are generated during union formation when required, and union formation can therefore add an individual to or remove an individual from the simulated population; the model does not maintain a separate persistent external population.

NetLogo patches and household coordinates are used only for graphical display. Patch state, geographic distance, neighbourhood relationships, and household coordinates do not affect demographic or resource-regulation processes.

2.2 State variables

Table 1 summarises the behaviourally relevant state variables and global model state. User-specified parameters and fixed structural constants are described under Initialisation, Input data, and the corresponding Submodels rather than being treated as dynamic state variables here. Purely auxiliary variables used transiently within procedures or only for interface display are likewise omitted unless they affect subsequent model behaviour.

The nominal carrying capacity (**carrying-capacity**), density-effect scaling and steepness parameters, demographic-schedule parameters, household-initialisation parameters, residence rule, and labour demand per capita are run-level parameters or structural constants rather than dynamic state variables. The mathematical quantity q introduced in Section 7.7 is likewise a derived effective density load rather than a separate entity state unless explicitly stored by a particular implementation.

In every density-regulated version, **densityEffect** is stored as a global model variable and updated before individual mortality trials. Its calculation depends on temporal resolution and resource-regulation formulation: annual versions store the density-dependent increment calculated for the current annual update, whereas daily versions convert the corresponding annual-equivalent density effect to a daily probability before storing it. The alternative definitions of demographic load and usable capacity entering this calculation are described in Section 7.7.

2.3 Temporal, spatial, and population scales

The annual model series (v1.1–v1.2.2) uses a discrete temporal resolution of one year. Household and individual ages are stored in units of years and the principal demographic event trials are evaluated once per annual update. The daily series (v1.3–v1.4.2) uses a discrete temporal resolution of one day, assumes a fixed year length of 365 days, and stores household and individual ages in units of days while indexing age-specific demographic schedules by completed years.

The model has no intrinsic temporal extent. Simulation duration is specified by the experimental configuration or interactive run settings. Technical stopping

conditions such as a maximum number of iterations therefore define experiment duration rather than the temporal scale of the demographic system.

The model has no behavioural spatial resolution or spatial extent. Households occupy NetLogo coordinates for visualisation, but distance, neighbourhood, patch identity, and movement through geographic space do not affect demographic events. The simulated population instead has a conceptual membership boundary through which individuals can enter or leave when unions are formed with external partners.

Population size is endogenous rather than fixed. The initial number of households is supplied as a model parameter, after which births, deaths, and union-related boundary movements determine population size. The technical `max-population` stopping value used in experiments is not a demographic carrying capacity and is not part of the model’s ecological regulation mechanism.

2.4 Model versions

Table 2 identifies the temporal resolution and resource-regulation formulation of each version. The version structure is included here because it determines both the temporal scale and which state variables are present.

Here N denotes population head count, D age-weighted subsistence demand, K nominal carrying capacity, and K_{eff} labour-adjusted effective carrying capacity. These quantities and their role in mortality regulation are defined in Section 7.7.

2.5 Rationale

The household is represented as the explicit NetLogo agent because co-residence, partnership, descent, household continuity, post-marital residence, and household fission operate at that level. Individual demographic state is retained through indexed member records because births, deaths, fertility, union eligibility, and reproductive state operate at the person level. This nested representation retains the individual heterogeneity and relational information required by the model without representing every person as a separate NetLogo agent.

The model is non-spatial because its focal mechanisms concern demographic schedules, household organisation, population composition, and population-level resource regulation rather than the spatial distribution of households or resources. Resource limitation is therefore represented through population-level quantities; explicit land use, production locations, environmental heterogeneity, and geographic movement belong to socioecological models coupled to the demographic kernel rather than to the Household Demography model itself.

3 Process overview and scheduling

Following initialisation, all model versions execute the same ordered demographic cycle at each time step. The temporal resolution and the implementation of reproduction and resource regulation differ between the annual and daily model

series, but the high-level ordering of demographic processes is retained. Details of the individual processes are provided in Section 7.

3.1 Common process sequence

Each time step proceeds through the following sequence:

1. increment household and individual ages;
2. calculate the population-level quantities required for mortality in the current model version and apply mortality to individual household members;
3. identify unpartnered individuals eligible for union formation, construct sex-specific candidate pools, form internal or external unions, and resolve the associated residence and household-fission processes;
4. apply fertility or reproductive-state updates to partnered women; and
5. dissolve households left without socially mature members and, where possible, reassign their dependent children to surviving households.

The ordering is consequential. Mortality precedes union formation, so an individual whose partner dies can become eligible for another union during the same time step. Union formation precedes fertility, so a newly formed couple can reproduce or initiate pregnancy during that time step. Dependent-child reassignment occurs only after the preceding demographic and household changes have been resolved.

3.2 Annual versions

In v1.1–v1.2.2, one iteration represents one year. Household and individual ages increase by one year at the beginning of each update. Mortality, union eligibility, and fertility are evaluated with annual age-specific probabilities. In the annual fertility process, a successful fertility trial produces a birth immediately within the same update.

In v1.2–v1.2.2, the population-level density effect is calculated before individual mortality trials. Version v1.2 calculates this effect from population head count relative to nominal carrying capacity; v1.2.1 uses age-weighted subsistence demand in place of head count; and v1.2.2 additionally calculates available labour and labour-adjusted effective carrying capacity. The resulting annual `densityEffect` is then used throughout that mortality update.

3.3 Daily versions

In v1.3–v1.4.2, one iteration represents one day. Household and individual ages increase in days, while age-specific fertility, union-formation, and baseline mortality schedules continue to be indexed by completed age in years. Annual

event probabilities are converted to corresponding daily probabilities as described in Section 7.10.

Daily reproduction retains the position of fertility in the common process sequence but replaces immediate birth with explicit reproductive states. For each household member, the postpartum-amenorrhoea countdown is first reduced by one day when positive. An active gestation countdown is then reduced by one day; when it reaches zero, a newborn is added and the mother's postpartum-amenorrhoea countdown is set to **amenorrhea-period-in-days**. After these state updates, a partnered woman whose gestation and amenorrhoea values are both zero is tested for pregnancy initiation using the daily fertility probability. A successful trial sets **hh_membersGestation** to **gestation-period-in-days**.

In v1.4–v1.4.2, density-dependent mortality is evaluated at each daily update. Version v1.4 uses current population head count; v1.4.1 recalculates current age-weighted subsistence demand; and v1.4.2 recalculates subsistence demand, available labour, labour requirements, and effective carrying capacity. The corresponding annual-equivalent density effect is converted to a daily probability and stored in **densityEffect** before individual mortality trials.

3.4 Scheduling within processes

Household-level processes use NetLogo **ask households** blocks, so households are processed in the stochastic agent order generated by NetLogo for each such call rather than according to a fixed household identifier. Within a household, individual records are processed through their list indexes according to the ordering specified by the corresponding procedure.

Union formation contains an additional order-dependent matching stage. Candidate pools **womenToMarry** and **menToMarry** are rebuilt at each update by evaluating unpartnered individuals against their age- and sex-specific union-formation probabilities. Because households are visited through **ask households**, the order in which candidates enter these lists inherits stochastic household visitation. The matching procedure then iterates through **womenToMarry** and searches **menToMarry** sequentially for the first acceptable internal partner. Once a match is formed, that male candidate is removed from the pool. A woman for whom no acceptable internal match remains forms a union with an external partner; after all female candidates have been processed, any remaining male candidates likewise form external unions. Matching outcomes can therefore depend on the realised ordering of otherwise eligible candidates.

Household membership changes caused by union formation and household fission are resolved before the fertility or reproductive-state update. Mortality-related deletions and partnership-status changes are resolved before union candidate pools are constructed, and all these processes are completed before dependent-child reassignment.

3.5 Rationale

The fixed ordering of high-level processes makes within-step consequences explicit: death can change subsequent union eligibility, union formation can change subsequent reproductive eligibility and household membership, and these changes can determine which households remain viable at the end of the update. The daily model series preserves this causal ordering while distributing demographic events across days and introducing gestation and postpartum amenorrhoea as explicit states. This allows temporal resolution to be varied without redefining the common demographic and household architecture.

4 Design concepts

4.1 Basic principles

The model combines demographic microsimulation with household-level social organisation and, in selected versions, population-level density-dependent mortality. Population change is generated from discrete births, deaths, and movements across the population boundary rather than from an imposed aggregate growth equation. Fertility and union formation use flexible Peristera–Kostaki age-specific probability schedules [Peristera and Kostaki, 2007, 2015], while baseline mortality uses age- and sex-specific Coale–Demeny model life tables [Coale et al., 1983, Jones, 2007]. Household structure mediates co-residence, partnership, descent, post-marital residence, household fission, and dependent-child reassignment.

The density-regulated versions represent resource limitation as a negative demographic feedback acting through mortality. Nominal carrying capacity is not a hard population ceiling. Successive versions replace simple head count with age-weighted subsistence demand and then condition the realisable fraction of nominal carrying capacity on age-weighted labour supply. These mechanisms provide a schematic link between demographic composition and resource regulation without explicitly representing production, food allocation, nutrition, disease transmission, land use, or environmental heterogeneity.

4.2 Emergence

Population trajectories, population growth or decline, persistence or extinction, age and sex structure, household number, household size and composition, household genealogies, and the realised timing and frequency of demographic events emerge from the sequence of stochastic individual and household events. No aggregate population-growth trajectory or target population structure is imposed.

In regulated versions, population-level subsistence demand, available labour, effective carrying capacity, and density-dependent mortality pressure are recalculated from the current population composition. These quantities are therefore derived from the emergent demographic state rather than supplied as external time series. By contrast, demographic probability schedules, household rules,

resource-regulation functions, structural constants, and process ordering are imposed model components.

4.3 Adaptation

Adaptation is not represented in the ODD sense of entities changing behaviour or strategies in response to their performance or previous outcomes. Household membership and demographic state change through mortality, union formation, reproduction, residence, fission, and dependent-child reassignment, but the rules governing these processes do not adapt during a simulation. Resource stress changes mortality probabilities rather than inducing behavioural responses such as altered production, technological change, fertility control, or general migration.

4.4 Objectives

Neither households nor individual records have an explicit objective function. Union formation, residence, household fission, reproduction, mortality, and dependent-child reassignment follow stochastic demographic schedules and fixed rules rather than optimisation of survival, reproduction, household size, resource acquisition, or another measure of performance.

4.5 Learning

No learning process is represented. Previous experience does not alter demographic probabilities, household rules, residence rules, or resource-regulation mechanisms during a run. Variables retaining historical information, such as household lineage and household age, constitute model state rather than learned information.

4.6 Prediction

Households and individual records do not estimate or forecast future demographic, social, or resource conditions. Model processes depend on current state variables, fixed probability schedules, and fixed social and resource-regulation rules. Gestation and postpartum-amenorrhoea countdowns represent ongoing physiological state rather than prediction of future events.

4.7 Sensing

The model contains no cognitive perception process. In ODD terms, the information required by each rule is accessed directly and without observational error from model state. Union formation uses individual age, sex, partnership status, household lineage, the current candidate pools, and the selected residence-descent rule. Mortality uses individual age and sex together with the current population-level density effect in regulated versions. Household fission uses the receiving household's current number of couples and its maximum couple

capacity. No rule depends on spatial neighbourhoods, geographic distance, or imperfect information.

4.8 Interaction

Interaction occurs principally through union formation and household membership. Eligible female and male records can be paired across households, subject to the lineage-distance restriction. The residence-descent rule determines which partner changes household membership, and formation of an additional couple can trigger household fission. Members of the same household also affect one another indirectly because births, deaths, unions, and fission alter household composition and subsequent demographic context. Dependent children from households without socially mature members can be reassigned to surviving households.

Internal union matching is population-wide rather than spatially local: candidate partners are drawn from the current global candidate pools and filtered by lineage relationship rather than geographic distance. In density-regulated versions, individuals additionally interact indirectly through population-level feedback, because their combined head count, age-weighted demand, and, where applicable, labour supply affect the mortality increment experienced throughout the population.

4.9 Stochasticity

Stochasticity enters initialisation through the generation of household ages, household couple capacities, founding couples, and reconstructed reproductive histories. During simulation, stochastic processes include individual mortality trials, union-eligibility trials, fertility or pregnancy-initiation trials, the generation of external partners, sex assignment at birth, and reassignment of dependent children to surviving households. Union formation also inherits stochasticity from NetLogo’s random household visitation order when candidate pools are constructed, which can affect the order in which otherwise eligible individuals are considered for matching. All stochastic processes use the NetLogo random-number generator and can be reproduced from a specified random seed.

4.10 Collectives

Households are explicit collectives and also constitute the model’s NetLogo agents. Each household contains multiple individual demographic records, and its membership changes through birth, death, union-related movement, household fission, and dependent-child reassignment. Household state and membership can consequently affect later processes experienced by its members.

Household lineages are represented through inherited identifiers that record descent relationships among households, but lineages are not independent agents or behavioural collectives. The total population is likewise an aggregate of household members rather than an agent with its own decision rules.

4.11 Observation

The model reports population and household counts, sex-specific population totals and age structures, births, deaths, union-related movements across the population boundary, and other event counters. Resource-regulated versions additionally expose age-weighted subsistence demand, available and required labour, the realised labour share, nominal carrying capacity, effective carrying capacity, and the density effect where applicable. Household and member-level records, including lineage information, can also be exported for inspection and analysis.

Quantities used only in the accompanying analysis, including outcome classifications, population-pressure regimes, sensitivity-analysis results, random-forest importance measures, partial-dependence estimates, and other statistical summaries, are calculated outside the model and are not model state or model-generated behaviour.

4.12 Rationale

The design separates demographic events, household organisation, and resource feedback so that each can be varied while retaining a common model architecture. Processes are represented only at the level required to examine how demographic schedules, household rules, population composition, temporal resolution, and resource limitation interact. Cognitive decision-making, adaptive subsistence behaviour, explicit spatial processes, and detailed production systems are excluded because they are outside the purpose of the demographic kernel and can instead be introduced by models coupled to it.

5 Initialisation

Initialisation creates a stochastic household population from user- or experiment-specified parameter values. The NetLogo random-number-generator seed can be controlled so that both initialisation and subsequent stochastic events are reproducible.

For each initial household, **hh.lineage** is initialised as a one-element list containing that household's NetLogo identifier. Household age, **hh.age**, is drawn uniformly from the inclusive bounds supplied by **household-initial-age-distribution**, and maximum couple capacity, **hh.maxCoupleCount**, is drawn uniformly from the inclusive bounds supplied by **max-couple-count-distribution**. In daily versions, household age is first sampled in whole years and then assigned a uniformly random additional number of days within the year.

Each household is founded by one female and one male, both assigned to couple index 0. For each founder, an age at union formation is generated stochastically from the corresponding sex-specific union-formation schedule by testing successive ages from age 0 upwards. If the search reaches the implementation cut-off at 100 years, the corresponding sex-specific modal-age parameter, **mu-women** or **mu-men**, is used instead. The founder's current age is then obtained by adding

the household age to the generated age at union formation. In annual versions, founder ages are capped at 100 years; daily versions store the resulting age in days and do not apply this cap.

5.1 Annual initialisation

In v1.1–v1.2.2, the previous reproductive history of each founding couple is reconstructed retrospectively in annual steps over the lifetime of the household. Beginning with the couple at its generated union age, the model evaluates the female fertility schedule once for each previous year. Each successful fertility trial creates one offspring, whose initial age corresponds to the number of years between the reconstructed birth and the beginning of the simulation. Initial household composition therefore depends jointly on household age, founders’ generated union ages, the fertility schedule, and stochastic fertility trials.

5.2 Daily initialisation

In v1.3–v1.4.2, founder ages and household age are represented in days, and the household’s reproductive history is reconstructed retrospectively one day at a time. Founding members begin with `hh_membersGestation` = 0 and `hh_membersAmenorrhea` = 0. Daily fertility probabilities are obtained from the annual fertility schedule using the same annual-to-daily probability conversion used during simulation. Successful fertility trials initiate a gestation countdown; completion of gestation creates an offspring with the corresponding current age and initiates postpartum amenorrhoea. The retrospective reconstruction can consequently leave the founding woman in an active gestation or postpartum-amenorrhoea state at the beginning of the simulated run.

After household membership has been initialised, each household is placed on one of the least occupied NetLogo patches for visualisation. This placement has no effect on subsequent demographic behaviour.

Demographic probability tables are generated or selected from the current fertility, union-formation, mortality-region, and mortality-level parameters before they are required for household construction. Resource-regulated versions additionally initialise the parameters and population-level quantities required by their respective mortality-regulation mechanisms.

The initial number of households, household-age bounds, household couple-capacity bounds, demographic-schedule parameters, and resource-regulation parameters are run-level inputs rather than fixed properties of the model. The parameter ranges used in the experiments accompanying the article are therefore documented with the experimental design rather than as part of the ODD model definition.

Implementation note. During preparation of the final model documentation, an inconsistency was identified in the retrospective reproductive-history procedure used to initialise the daily model versions. In the implementation used for the simulation experiments reported in the accompanying article, gestation and

postpartum amenorrhoea were not updated using exactly the same sequence and eligibility conditions as during subsequent daily simulation. The released implementation was revised so that retrospective initialisation first advances postpartum amenorrhoea, then gestation and any resulting birth, and finally evaluates pregnancy initiation only when both reproductive-state counters are zero. This correction affects only construction of the initial household population; reproductive dynamics after simulation start were already implemented using the intended logic. Diagnostic comparison of populations generated at setup showed no substantial change in their age-sex structure. The simulation batches reported in the article were therefore retained. The released implementation was revised so that retrospective initialisation uses the same reproductive-state update sequence and age-specific pregnancy probability as the running daily model, evaluated at the woman’s reconstructed age on each historical day.

5.3 Rationale

The model does not initialise the population from a prescribed age pyramid or stable-population distribution. Instead, each initial household is assigned a stochastic history using the same demographic schedules that govern subsequent model dynamics. This generates heterogeneous household compositions while avoiding an externally imposed initial population structure. Initial demographic composition is consequently one source of stochastic variation among replicate runs.

6 Input data

The model uses static external reference data for baseline mortality but no time-varying external input data during a simulation run. Fertility, union formation, household organisation, and resource-regulation dynamics are generated from model state, stochastic processes, fixed structural rules, and run-level parameter values rather than being driven by empirical time series or spatial datasets.

Baseline mortality is supplied through age- and sex-specific Coale–Demeny model-life-table vectors [Coale et al., 1983]. The release includes separate mortality-table files for each combination of regional family and sex in the `demotables` subdirectory. These files were generated in R from the Coale–Demeny data distributed through the *demogR* package [Jones, 2007]. At setup, the selected Coale–Demeny region and mortality level determine which female and male schedules are retrieved from these files and stored as the mortality lookup vectors used during the simulation. The selected baseline vectors remain fixed throughout the run. In density-regulated versions, the current density-dependent mortality increment is calculated separately and combined with the corresponding baseline mortality probability when mortality is evaluated.

Fertility and union-formation schedules are generated internally from the Peristera–Kostaki parameters specified for the run and therefore constitute parametrised submodels rather than external input data. Likewise, carrying

capacity, density-effect parameters, labour demand, initial household conditions, residence rules, and other user- or experiment-specified quantities are model parameters rather than input datasets.

The core model requires no GIS layers, weather or climate series, crop-production data, historical census series, archaeological population reconstructions, or other external environmental or demographic time series. Such inputs can instead be introduced by socioecological models coupled to the Household Demography kernel.

6.1 Rationale

Bundled mortality tables provide a standardised empirical demographic reference while keeping the runtime model independent of external services or changing data sources. The absence of time-varying external drivers is deliberate: the model isolates the consequences of demographic schedules, household organisation, stochastic life events, temporal resolution, and resource-regulation mechanisms without tying simulation dynamics to a particular historical population or environmental sequence.

7 Submodels

7.1 Parametric fertility and union-formation schedules

Fertility and union formation use the same flexible parametric form as Model 1 in Peristera and Kostaki [Peristera and Kostaki, 2007, 2015]. For age x ,

$$\sigma(x) = \begin{cases} \sigma_1, & x \leq \mu, \\ \sigma_2, & x > \mu, \end{cases} \quad (1a)$$

$$f_{PK}(x; c_1, \mu, \sigma_1, \sigma_2) = c_1 \exp \left[- \left(\frac{x - \mu}{\sigma(x)} \right)^2 \right]. \quad (1b)$$

Here c_1 is the maximum event probability, μ is the modal age, and σ_1 and σ_2 control spread before and after the mode. The function is interpreted as an age-specific event-probability schedule rather than as a normalised probability distribution.

We use separate parameterisations for fertility, female union formation, and male union formation. In annual versions, the corresponding values are annual event probabilities. In daily versions, they are converted as described in Section 7.10.

At setup, the model evaluates each parameterisation at integer ages from 0 to 150 years, producing one fertility lookup vector and separate female and male union-formation vectors. Ages beyond the represented range receive an event probability of zero. In annual versions, these lookup values are used directly. In daily versions, age in days is first converted to completed years and the

corresponding annual probability is then converted to a daily probability as described in Section 7.10.

7.2 Union eligibility, matching, and boundary movement

At each update, currently unpartnered individuals are evaluated against their sex-specific union-formation schedule. For sex s and age x , the annual eligibility probability is

$$P_{\text{union},y}(x \mid s) = f_{\text{PK}}(x; \theta_s). \quad (2)$$

Individuals passing the stochastic eligibility test are appended to sex-specific candidate pools. Households contribute candidates in the stochastic visitation order generated when the household agentset is processed, while member records within each household are traversed in stored index order.

Matching is sequential rather than simultaneous. Female candidates are processed in candidate-list order. For each female candidate, the model scans the current male candidate pool from its beginning and forms a union with the first male whose household satisfies the lineage-distance restriction. Once matched, that male is removed from the candidate pool. If no acceptable internal male remains, the female candidate forms a union with an external partner. After all female candidates have been processed, any remaining male candidates likewise form external unions. Matching outcomes can therefore depend on candidate ordering.

External unions generate boundary movement according to the composite residence–descent rule. Under matrilineal–matrilocal residence, an external male enters when the resident partner is female, whereas an eligible resident male leaves the simulated population. Under patrilineal–patrilocal residence, an external female enters when the resident partner is male, whereas an eligible resident female leaves. Only the focal resident individual is removed in an outflow event; dependent children do not automatically accompany that individual. These exchanges constitute union-related boundary movement rather than a general migration process.

When an external partner enters the population, the partner has the opposite represented sex and is assigned an age using the sex-specific union-formation schedule. In daily versions, an entering partner is initialised with neither an active gestation nor postpartum amenorrhoea.

The model does not include divorce, intentional fertility control, multiple simultaneous partners, or unions other than the female–male pairings represented by the current matching algorithm.

7.3 Lineage restriction, post-marital residence, and household fission

Each household records its descent sequence through household fission. For two households A and B , let c denote the length of the shared prefix of their lineage

identifiers. Lineage distance is

$$d(A, B) = (|A| - c) + (|B| - c). \quad (3)$$

A potential internal union is acceptable when

$$d(A, B) > \tau_{\text{kin}}, \quad (4)$$

where τ_{kin} is the parameter **acceptable-kinship-degree-for-couples**. Larger values of d indicate more distant household lineages, so this parameter is a minimum acceptable lineage distance rather than a coefficient of relatedness.

The model provides two composite residence-descent configurations: matrilineal-matrilocal and patrilineal-patrilocal. Under the former, the male partner joins the female partner’s household; under the latter, the female partner joins the male partner’s household. Descent and locality change together in the present model family and their effects are therefore not separable.

A receiving household can contain no more than its stored maximum number of co-resident couples. When a new couple would exceed this limit, the model creates a daughter household. The daughter household inherits the parent’s couple capacity, extends its lineage identifier, starts with household age zero, and initially contains only the newly formed couple. Existing individuals retain their demographic state when transferred between households; in daily versions this includes any active gestation or postpartum-amenorrhoea state. Removal from households of origin is deferred until the union-matching procedure has been completed so that member indexes remain stable during matching. Spatial relocation of the new household in the NetLogo world is for visualisation only.

7.4 Fertility and reproduction

In annual versions, fertility applies only to currently partnered women. For a partnered woman of age x , the probability of giving birth during an annual update is

$$P_{\text{birth},y}(x) = f_{\text{PK}}(x; \theta_f). \quad (5)$$

A successful event adds one newborn to the couple’s household. Newborn sex is assigned with probability 0.5 for each of the two represented sexes.

In daily versions, the same fertility schedule governs pregnancy initiation rather than immediate birth. The detailed conversion and reproductive-state rules are described in Section 7.10.

The model does not include intentional fertility control, multiple births from a single pregnancy, pregnancy-specific maternal mortality, or parity-dependent fertility. In daily versions, pregnancy and postpartum amenorrhoea affect subsequent reproductive eligibility only through the explicit reproductive-state variables described in Section 7.10.

7.5 Baseline mortality

Baseline mortality is represented with annual age- and sex-specific probabilities derived from the Coale–Demeny regional model life tables [Coale et al., 1983]. At setup, separate female and male lookup vectors are loaded from the bundled files in the `demoTables` directory. These files were generated from the Coale–Demeny tables using the `demogR` package [Jones, 2007] and the R script distributed with the model. The selected regional family determines the pair of sex-specific files and the selected mortality level determines the column extracted from each file. For age x , sex s , regional family r , and mortality level ℓ ,

$$P_{\text{death},y}(x \mid s, r, \ell) = f_{\text{CD}}(x, s; r, \ell). \quad (6)$$

Mortality is evaluated independently for each individual. A death removes the individual from the household and population and updates any partnership link affected by that death. In density-independent versions, this baseline probability is used directly. Density-regulated versions add the increment described in Section 7.7.

7.6 Household viability and dependent-child reassignment

At the end of each demographic update, households containing no socially mature members are dissolved. Social maturity is defined as age 15 years, or the equivalent age in days in daily versions. Remaining dependent individuals are pooled and independently assigned to randomly selected surviving households. Reassignment preserves sex and age but resets partnership status to unpartnered; daily versions additionally initialise gestation and postpartum amenorrhoea to zero. Thus, the household-viability rule treats reassigned dependants as socially dependent irrespective of any union or reproductive state that could arise probabilistically below the social-maturity threshold. If no receiving household survives, reassignment is impossible and the population is extinct.

7.7 Density-dependent mortality

Density-regulated versions add a population-level mortality increment to the baseline age- and sex-specific mortality probability. Let q denote the effective density load, α the parameter `density-effect-scaling-factor`, and η the `density-effect-steepness`. The annual-equivalent density increment is

$$g(q) = \alpha q^\eta. \quad (7)$$

The scaling parameter α is therefore the density-dependent mortality increment when $q = 1$; it is not an upper bound on the unbounded function $g(q)$. In the NetLogo implementation, a constant of 10^{-6} is added to the capacity denominator as a numerical safeguard against division by zero. This safeguard does not alter the conceptual definition of q given below.

The experiments reported in the associated article use $\eta = 3$ as a fixed structural constant. The annual net mortality probability is

$$P'_{\text{death},y}(x | s) = \text{clamp}_{[0,1]} [P_{\text{death},y}(x | s) + g(q)], \quad (8)$$

where

$$\text{clamp}_{[0,1]}(z) = \min\{1, \max\{0, z\}\}. \quad (9)$$

The definition of q differs by version:

$$q = \begin{cases} N/K, & \text{v1.2 and v1.4,} \\ D/K, & \text{v1.2.1 and v1.4.1,} \\ D/K_{\text{eff}}, & \text{v1.2.2 and v1.4.2.} \end{cases} \quad (10)$$

Thus, density-dependent mortality always retains the same nonlinear functional form while successive model versions alter the demographic load or usable capacity entering that function.

7.8 Age-weighted subsistence demand

Versions v1.2.1, v1.2.2, v1.4.1, and v1.4.2 represent resource demand in adult-equivalent units. The demand weight of an individual of age x is

$$k(x) = \begin{cases} \left(\frac{x}{x_m}\right)^{1.2}, & x < x_m, \\ 1, & x_m \leq x \leq x_s, \\ \left(\frac{x_s}{x}\right)^2, & x > x_s, \end{cases} \quad (11)$$

with fixed breakpoints $x_m = 15$ and $x_s = 45$ years. These breakpoints and exponents are schematic model constants, not empirically calibrated biological thresholds. In particular, the numerical equality between x_m and the social-maturity threshold does not imply that the two quantities have the same conceptual role.

Total demand is

$$D = \sum_{i=1}^N k(x_i). \quad (12)$$

Replacing head count with D allows populations with identical numbers of individuals but different age structures to exert different demographic loads.

7.9 Age-weighted labour and effective capacity

Versions v1.2.2 and v1.4.2 additionally represent age-dependent labour supply. The labour contribution of an individual of age x is

$$l(x) = \begin{cases} \left(\frac{x}{x_m}\right)^{10}, & x < x_m, \\ 1, & x_m \leq x \leq x_s, \\ \left(\frac{x_s}{x}\right)^{10}, & x > x_s, \end{cases} \quad (13)$$

using the same breakpoints $x_m = 15$ and $x_s = 45$ but a steeper decline outside the central adult-age interval. Available labour is

$$L_a = \sum_{i=1}^N l(x_i). \quad (14)$$

Let λ be **labour-demand-per-capita**. The labour required to realise the nominal environmental capacity is

$$L_K = K\lambda. \quad (15)$$

The effective capacity is then

$$K_{\text{eff}} = K \min\left(1, \frac{L_a}{K\lambda}\right). \quad (16)$$

Effective capacity can therefore equal but never exceed nominal capacity, while falling below it when the population lacks sufficient age-weighted labour. The mortality mechanism then uses $q = D/K_{\text{eff}}$.

The demand and labour functions do not distinguish sex, activity, season, health, household technology, or task specialisation. They are deliberately schematic links between demographic composition and resource regulation.

In daily versions, individual age in days is converted to completed years before evaluation of the demand and labour weighting functions, so both weights change at yearly age boundaries rather than continuously between birthdays.

7.10 Conversion from annual to daily probabilities

In v1.3–v1.4.2, an annual event probability P_y is converted to the corresponding daily probability by

$$\mathcal{C}_n(P_y) = 1 - (1 - P_y)^{1/n}, \quad (17)$$

with $n = 365$. For a fixed annual event probability, repeated independent daily trials using this conversion reproduce the corresponding cumulative annual probability.

For daily fertility, the probability of initiating pregnancy at age in days x_d is

$$P_{\text{preg},d}(x_d) = \mathcal{C}_{365} [f_{\text{PK}}(\lfloor x_d/365 \rfloor; \theta_f)]. \quad (18)$$

A successful event starts a gestation countdown fixed at 280 days. When it reaches zero, a birth is generated and the mother's postpartum-amenorrhoea

countdown is set to the parameter `amenorrhea-period-in-days`. A new pregnancy can begin only when the woman is partnered, not currently gestating, and not in postpartum amenorrhoea.

For daily union formation, the probability that an unpartnered individual of sex s and age in days x_d becomes eligible is

$$P_{\text{union},d}(x_d | s) = \mathcal{C}_{365} [f_{\text{PK}}(\lfloor x_d/365 \rfloor; \theta_s)]. \quad (19)$$

For daily density-dependent mortality, the annual-equivalent density increment is bounded before conversion:

$$g_d(q) = \mathcal{C}_{365} \left[\text{clamp}_{[0,1]}(g(q)) \right]. \quad (20)$$

Daily mortality is

$$P'_{\text{death},d}(x | s) = \text{clamp}_{[0,1]} \left[\mathcal{C}_{365}(P_{\text{death},y}(x | s)) + g_d(q) \right]. \quad (21)$$

In demand- and labour-weighted daily versions, the quantities entering q are recalculated during daily mortality updates.

7.11 Code correspondence

To facilitate comparison between this ODD and the NetLogo implementation, Table 3 lists the principal procedures documented in the manuscript. Procedure names are retained from the code even where descriptive text uses the broader term “union formation” instead of “nuptiality”.

The model is implemented in NetLogo 7.0.4 [Wilensky, 1999].

7.12 Implementation corrections relative to the reported simulations

While preparing the final model documentation, we identified two minor inconsistencies in the daily and union-related implementation used for the simulation experiments reported in the accompanying article.

First, the retrospective reproductive-history procedure used to initialise the daily model versions did not update gestation and postpartum amenorrhoea using exactly the same sequence and eligibility conditions as the reproductive procedure used during subsequent simulation. The released implementation was revised so that retrospective initialisation advances postpartum amenorrhoea, then gestation and any resulting birth, and evaluates pregnancy initiation only when both reproductive-state counters are zero and using the woman’s reconstructed age.

Second, the procedure used to generate ages for founding and externally entering partners incremented the candidate age after each stochastic union-formation trial, including successful trials. This produced an unintended one-year upward offset in the generated age and, at the upper search boundary, could trigger the fallback to the modal union age. The released implementation was

revised so that a successful stochastic trial returns the age at which that trial occurred.

These corrections affect initial population construction and the ages assigned to externally entering partners, but do not alter the demographic schedules, household rules, or subsequent fertility and mortality mechanisms. Diagnostic comparison of setup populations showed no substantial change in age-sex structure, and comparison of a sample of trajectories across model versions showed no detectable change in the resulting dynamics. We therefore retained the simulation batches reported in the article. The corrections are incorporated in the released source code and documented in the repository version history.

Third, the daily dependent-child reassignment procedure reset gestation and postpartum-amenorrhoea counters when individuals were transferred from a dissolved household to a surviving household. This behaviour was unintended. The released implementation preserves these individual reproductive-state variables during reassignment while continuing to reset partnership status, consistent with the model’s treatment of reassigned dependants as individually adopted members of the receiving household. The affected state combination is expected to be rare because household dissolution requires all remaining members to be younger than the fixed social-maturity threshold of 15 years, while union formation and fertility at these ages occur only probabilistically under parameterisations with sufficiently early demographic schedules.

7.13 Rationale

The submodels are intentionally modular. Fertility, union formation, mortality, household organisation, resource demand, labour supply, and temporal scheduling can therefore be examined or replaced without rewriting the model as one monolithic demographic process. The eight model versions use this modularity as controlled structural experiments: each adds or changes a limited set of mechanisms while retaining the common household representation and demographic schedules.

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Table 1: Behaviourally relevant entity and global state in the Household Demography model. Code-level names refer to the current implementation.

Entity or level	State variable	Meaning
Household	identifier (who)	Unique NetLogo identifier of the household; household identifiers are also used within lineage histories.
Household	lineage (hh_lineage)	Ordered list representing the household's descent history and used to calculate lineage distance between households.
Household	household age (hh_age)	Time elapsed since creation of the household; expressed in years in annual versions and days in daily versions.
Household	maximum couple capacity (hh_maxCoupleCount)	Maximum number of co-resident couples accepted by the household before formation of an additional couple causes household fission.
Individual record	age (hh_membersAge)	Age of each household member; expressed in years in annual versions and days in daily versions.
Individual record	sex (hh_membersSex)	Boolean sex state; true represents female and false represents male in the current implementation.
Individual record	partnership index (hh_membersMarriage)	Integer identifying the couple to which a household member belongs; -1 denotes an unpartnered individual.
Individual record, daily versions	gestation countdown (hh_membersGestation)	Non-negative integer countdown representing an ongoing pregnancy. A value of 0 denotes no active gestation; pregnancy initiation sets the value to gestation-period-in-days , after which it decreases by one per daily update until reaching 0, when birth occurs.
Individual record, daily versions	postpartum amenorrhoea countdown (hh_membersAmenorrhea)	Non-negative integer countdown representing postpartum amenorrhoea. A value of 0 denotes no active amenorrhoea; following birth, the value is set to amenorrhea-period-in-days and decreases by one per daily update until reaching 0.
Population/global	demographic probability tables (fertilityTable , nuptialityTable-women , nuptialityTable-men , mortalityTable-women , mortalityTable-men)	Age-specific probability lookup tables constructed or selected from the demographic parameters at initialisation.
Population/global	union candidate pools (womenToMarry , menToMarry)	Temporary lists identifying currently eligible female and male candidates during union formation.
Population/global	dependent-child list (orphanList)	Temporary list of dependent children awaiting reassignment after their household is left without socially mature members.
Population/global, daily versions	time state (currentYear , currentDayOfYear)	Current simulated year and day within the year.
Population/global, demand-weighted	subsistence demand (carrying-capacity-demand)	Current total age-weighted demand expressed in adult-equivalent units.

Table 2: Model versions covered by this ODD.

Version	Time step	Resource regulation	Principal addition
v1.1	annual	none	density-independent demographic baseline
v1.2	annual	N/K	head-count-based density-dependent mortality
v1.2.1	annual	D/K	age-weighted subsistence demand
v1.2.2	annual	D/K_{eff}	labour-adjusted effective capacity
v1.3	daily	none	gestation, postpartum amenorrhoea, and daily event scheduling
v1.4	daily	N/K	daily head-count-based density regulation
v1.4.1	daily	D/K	daily age-weighted demand regulation
v1.4.2	daily	D/K_{eff}	daily demand and labour coupling

Table 3: Principal correspondence between ODD submodels and documented NetLogo procedures.

ODD process	Principal documented procedures or reporters
Annual demographic update	<code>update-households</code>
Mortality	<code>apply-mortality</code> ; <code>hh.update-members-survival</code> ; <code>hh.whichMembersDying</code> ; <code>build-mortality-tables</code> ; <code>load-coale-demeny-table</code> ; <code>get-mortality-prob</code> ; <code>get-mortality-prob-of-day</code>
Union candidate selection	<code>apply-nuptiality</code> ; <code>build-lists-to-marry</code> ; <code>hh.set-members-to-marry</code>
Union matching and kinship	<code>try-to-form-couples</code> ; <code>couple-is-acceptable</code> ; <code>get-kinship-degree</code>
Residence and external unions	<code>create-couple</code> ; <code>create-couple-external</code>
Household fission	<code>hh.try-to-add-couple</code> ; <code>hh.household-fission</code>
Fertility / reproduction	<code>apply-fertility</code> ; <code>hh.reproduce</code>
Dependent-child reassignment	<code>manage-orphanhood</code> ; <code>hh.dissolve-if-no-adults</code> ; <code>hh.add-orphan</code>
Density effect	<code>get-density-effect</code> ; <code>get-density-effect-of-day</code>
Age-weighted demand	<code>hh.get-total-carrying-capacity-demand</code> ; <code>get-carrying-capacity-demand-at-age</code>
Age-weighted labour and effective capacity	<code>hh.get-total-labour</code> ; <code>get-labour-at-age</code>